

ARCHITECTURE RECOMMENDATION

call_summarization

Post-call summarisation for a contact centre: summary, intents, commitments and compliance flags posted to the CRM within a minute of call end.

Recommended architecture	Prompt chain (sequential workflow) (workflow)
Budget verdict	Fits the latency budget
Estimated p50 latency	~26 s of a 60 s budget
Estimated cost per request	\$0.045 at list prices, no caching
Tokens vs. one model call	2x (4 model calls)
Score	+3.2 (fit +2.5, latency +1.0, cost -0.2)
Currently implemented	Single model call (changes)

Recommended: **Prompt chain (sequential workflow)** (score +3.2; est. p50 latency ~26s vs budget 60s, ~2x tokens of one call). The repository currently implements **Single model call** (score +0.2, ~9s); the recommendation changes the topology.

1. Why this architecture

- **+2.5** Steps are predictable: a code-defined chain with gates beats an agent deciding the path (more predictable, cheaper, easier to test). [Anthropic, Building effective agents]

2. Repository scan

8 files scanned under /Users/saxena/Desktop/ClaudeCode_GIT/AgentTopologyBuilder/case_study/01_call_summarization. Frameworks: anthropic-sdk. Tools found: 0. Knowledge sources: none. Side effects: external-post (irreversible), ticket/crm (irreversible). Current topology as implemented: **single_call** (Model calls without tools or loops.).

Current vs. recommended

	Current: Single model call	Recommended: Prompt chain (sequential workflow)
Score	+0.2	+3.2
Est. latency	~9 s	~26 s
Est. cost / request	\$0.070	\$0.045
Tokens vs. one call	1x	2x

Inferred profile fields

Field	Value	Conf.	Reason
tool_count	0	50%	No tool definitions found.
tool_calls_per_task	0	60%	No tools found.
tool_side_effects	irreversible	70%	Side-effecting integrations: external-post (irreversible), ticket/crm (irreversible)
knowledge_sources	0	50%	No retrieval integrations found.
retrieval_depth	none	50%	Derived from sources (0) and whether a loop can issue follow-up queries.
context_tokens_per_task	2000	30%	Rough function of sources and tools; replace with measured input tokens per request.
scope_breadth	1	40%	1 file(s) define system prompts/instructions; routing/handoffs raise this.
parallel_subtasks	1	30%	No orchestration detected.

Field	Value	Conf.	Reason
steps_predictable	True	50%	No run-time planning detected.
task_complexity	simple	30%	Defaulted from tool presence.
latency_budget_s	3600.0	50%	Batch/cron execution: minutes to hours are acceptable.
interaction	single_turn	50%	Batch jobs are single-shot.
verifiability	partial	50%	Tests or schema validation present.
error_recoverability	hard	50%	Irreversible integrations present.
accuracy_priority	4	40%	Irreversible actions raise the cost of errors.

Evidence (first hits per category)

Category	Label	Location	Snippet
side_effects	external-post (irreversible)	app/crm.py:10	requests.post("https://hooks.internal/compliance", json={"call_id": call_id, "flags": payl
side_effects	ticket/crm (irreversible)	app/crm.py:1	""Salesforce integration.""
side_effects	ticket/crm (irreversible)	app/crm.py:3	from simple_salesforce import Salesforce
frameworks	anthropic-sdk	app/pipeline.py:4	import anthropic
frameworks	anthropic-sdk	app/pipeline.py:11	client = anthropic.Anthropic()
serving	batch/cron	app/pipeline.py:5	from celery import Celery
quality	structured-output	app/pipeline.py:31	response = client.messages.parse(

3. Options compared

Score = structural fit + latency adjustment + cost adjustment. Latency is the estimated p50 on the critical path; cost sums every call at list prices. Non-viable options cannot do the job and are listed last.

#	Architecture	Score	Fit	Latency	Budget	Cost/req	Tokens	Calls
1	Prompt chain (sequential workflow)	+3.2	+2.5	~26 s	fits	\$0.045	2x	4
2	Single agent with tools	+1.5	+1.5	~14 s	fits	\$0.089	1x	2
3	Single model call	+0.2	+0.0	~9 s	fits	\$0.070	1x	1
4	Router / classifier dispatch	+0.2	+0.0	~10 s	fits	\$0.073	1x	2
5	Evaluator-optimizer loop	-0.2	+1.5	~44 s	tight	\$0.193	2x	6
6	Parallel voting / ensemble	-0.3	+1.0	~15 s	fits	\$0.274	4x	6
7	Orchestrator + worker subagents	-9.6	-9.0	~40 s	fits	\$0.082	2x	4
8	Parallel sectioning (code-defined fan-out) not viable: nothing to fan out	+1.0	+0.0	~17 s	fits	\$0.036	2x	5
9	Specialist handoffs (decentralised) not viable: no second specialist to hand off to	-4.8	-4.0	~28 s	fits	\$0.164	2x	5
10	Hierarchical teams (orchestrator -> leads -> workers) not viable: no domains or sub-task volume to justify two levels	-16.6	-13.0	~77 s	exceeds	\$0.242	8x	22

Latency/accuracy frontier (no option is both faster and a better structural fit): Single model call (~9 s, fit +0.0), Single agent with tools (~14 s, fit +1.5), Parallel voting / ensemble (~15 s, fit +2.5). Fastest viable within budget: **Single model call**. Strongest structural fit: **Parallel voting / ensemble**.

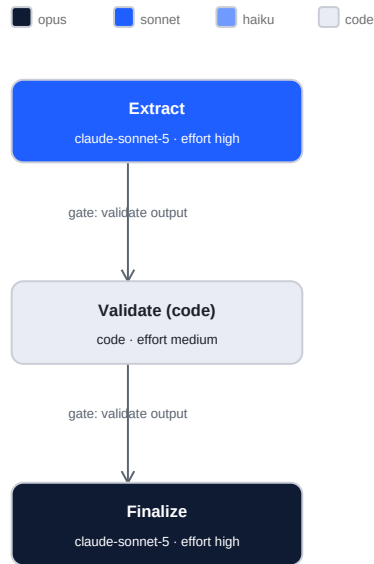
Runners-up

- **Single agent with tools** (score +1.5, ~14 s, 1x tokens): A 78% single-agent baseline is the regime where one agent plus a better model/effort/tools wins.

- **Single model call** (score +0.2, ~9 s, 1x tokens): Classification, extraction, summarisation, Q&A over a bounded context where the steps need no tools and no iteration..
- **Router / classifier dispatch** (score +0.2, ~10 s, 1x tokens): Routing easy requests to a small model is the cheapest accuracy-preserving lever.

4. Decomposition plan: Prompt chain (sequential workflow)

Fixed sequence of model steps defined in code, each consuming the previous output, with programmatic gates/checks between steps.



Component	Role	Model	Effort	Tools	Parallel
Extract	Pull candidate fields from the input (tools if needed).	claude-sonnet-5	high	-	-
Validate (code)	Schema and business-rule checks in code; reject or route back.	code	medium	-	-
Finalize	Resolve ambiguities and emit the final structured record.	claude-sonnet-5	high	-	-

5. Model selection per role

Catalog case-study-ecosystem; data class **confidential**; verified entries only. Providers used: anthropic. Estimated per request on the chosen models: ~25 s, \$0.059.

Role	Needs	Chosen	Effort	Fallback	Est./call	Alternatives
extract stage	tier ≥3; 30s; 12,000 ctx; tools; structured	claude-sonnet-5	high	claude-haiku-4-5	~9.0s \$0.0535	claude-haiku-4-5 (+1.4), claude-opus-5 (+0.2)
finalize synthesizer	tier ≥3; 30s; 5,700 ctx; structured	claude-sonnet-5	high	claude-haiku-4-5	~5.7s \$0.0154	claude-haiku-4-5 (+1.4), claude-opus-5 (+0.4)

- **extract:** Tier 4 vs required 3; est. 9.0s of a 30.0s share; \$0.0535 per call. measured extraction accuracy 90%
- **finalize:** Tier 4 vs required 3; est. 5.7s of a 30.0s share; \$0.0154 per call. measured extraction accuracy 90%

6. Agent report card (full)

Full report card for **call_summarization**: overall **B** (84.7/100). Task complexity 3 (moderate), design 1 (trivial), behaviour 2 (simple). Weakest dimension: Governance and guardrails (F).

Dimension	Grade	Score	Weight	Findings
Outcome quality	B	84.6	30	• Only 78% of repeated tasks agree on outcome across runs: reliability across runs is low (Galileo reports 60% single-run falling to 25% over eight runs).
Reliability and control	B	87.5	20	• High-stakes workload with little verification in traces and no eval harness in code (verification failures are ~25% of multi-agent failures).
Cost and token efficiency	A	100.0	15	-
Latency	A	100.0	15	-
Governance and guardrails	F	30.0	10	• No eval harness or tests found: build a ~20-query eval before changing structure. • Irreversible integrations without an approval or confirmation gate in code. • No tracing/observability library detected; multi-agent failures cannot be diagnosed without per-turn traces. • Traces show no human approval steps despite irreversible tools.
Complexity fit	B	88.0	10	• High stakes, little verification
Overall	B	84.7		

Complexity

Axis	Level
Task	3 (moderate)
Design	1 (trivial): 0 tools, delegation depth 0
Behaviour	2 (simple): steps p50 3.0, p95 3.0, CV 0.0, branching 1.0

Mismatches

- **High stakes, little verification** (warn): Only 0% of runs include a verification step for a stakes-4 workload.
Action: Add an evaluator/verifier step (tests, schema checks, or an independent model) before returning.

Per-agent cards

Agent	Role	Grade	Steps p50/p95	Tools/run	Errors	Loops	Tokens/run	Findings
agent	-	A	3.0/3.0	0.0	0%	0%	12,116	stable: no loops, low variance, no tool errors

7. Cross-cutting recommendations

- **Build a ~20-query eval before adding any structure.** Every source agrees: measure a single agent/call first, then add topology only where the eval shows it failing. Above a ~45% single-agent baseline, adding agents tended to hurt in the scaling study. Keep the eval to compare latency and accuracy across topologies. [Google/MIT, Towards a Science of Scaling Agent Systems]
- **Prompt caching on the stable prefix (system prompt, tool schemas, reference docs).** Multi-turn and multi-call topologies re-send the same prefix on every call; caching cuts cost up to ~90% and time-to-first-token. [Anthropic, Effective context engineering for AI agents]
- **Structured outputs (JSON schema) for routers, extractors and votes.** Removes parsing failures and makes code-side gates and majority votes trivial. [Anthropic, Building effective agents]
- **Dedicated, gated tools for irreversible actions with checkpoints/rollback.** Promote side-effecting actions to typed tools so the harness can intercept, confirm, audit and roll back; never let them run through a generic shell tool. [Anthropic, Building effective agents]

- **Consider a difficulty router in front.** Send easy requests to a small model and hard ones to the full path; the largest cost lever after caching. [Anthropic, Building effective agents]

8. Estimate assumptions

- 3 sequential stages with programmatic gates between them.
- Serving assumptions (reference model per capability level from the catalog): claude-opus-5 ~1.5s TTFT, ~55 tok/s; acme-onprem-70b ~0.5s TTFT, ~90 tok/s; claude-haiku-4-5 ~0.6s TTFT, ~140 tok/s; tool call ~0.5s.
- Reading load per task 9,000 tokens (context pressure: low); output type structured_data.
- Latency is the critical path (parallel branches count once); cost sums every call at list prices without caching. Treat both as order-of-magnitude.

Appendix A. Workload profile used

Field	Value	Field	Value
name	call_summarization	task_complexity	moderate
steps_predictable	True	scope_breadth	1
parallel_subtasks	1	output_type	structured_data
tool_count	0	tool_calls_per_task	1
tool_dependency	sequential	tool_side_effects	irreversible
tool_overlap	False	tool_latency_s	0.5
knowledge_sources	0	retrieval_depth	none
context_tokens_per_task	9000	accuracy_priority	4
single_agent_baseline	0.78	verifiability	strong
error_recoverability	moderate	latency_budget_s	60.0
interaction	single_turn	requests_per_day	50000
cost_sensitivity	4	human_in_loop	False
data_sensitivity	confidential	tools	[]
sources	[]		

Appendix B. Sources

- **ANTHROPIC_BEA:** Anthropic, Building effective agents
- **ANTHROPIC_CTX:** Anthropic, Effective context engineering for AI agents
- **ANTHROPIC_MAR:** Anthropic, How we built our multi-agent research system
- **COGNITION:** Cognition, Don't build multi-agents
- **OPENAI_PG:** OpenAI, A practical guide to building agents
- **SCALING:** Google/MIT, Towards a Science of Scaling Agent Systems

Full evidence and the consolidated decision framework: docs/industry_guidance.html and docs/METHODOLOGY.html in the VG Select: 3A repository.